Lezhi Wang

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EDUCATION

University of North Carolina at Chapel Hill — Chapel Hill, NC
Bachelor of Science in Statistics and Analytics, Minor in Data Science, GPA: 3.94 — Aug. 2023 – May 2027
Honors: Dean's List — Aug. 2023 – Present

SKILLS

Languages: *R, Python, SQL, Java, HTML/CSS, Latex*
Developer Tools: *VS Code, Jupyter Notebooks, Tableau, IntelliJ, GAMS, Excel, Jamovi, Git, Docker*
Libraries: *Tidyverse (dplyr, readr, ggplot2), Car, Leaps, MASS, pandas, NumPy, Matplotlib*
Coursework highlight:
Machine Learning, Data Models & Inference, Methods of Data Analysis, Linear Algebra, Probability, Discrete Mathematics.

WORKING EXPERIENCE

Undergraduate Independent Research — Aug. 2025 – Present
University of North Carolina at Chapel Hill — *Chapel Hill, NC*

- Designed a human-AI hybrid content generation model, representing outputs as a continuous mixture of human and LLM distributions
- Implemented game-theoretic simulation in Python (NumPy, Matplotlib) modeling 10,000+ agent market competition, achieving equilibrium convergence in <200 iterations via gradient descent optimization
- Conducted sensitivity analysis across 50+ parameter scenarios to analyze human-AI creator dynamics under varying structures
- Automated data visualization and reporting pipeline, documenting analytical findings in LaTeX technical reports for publication-track research

Data Analytics Intern — May 2024 – Aug. 2024
CITIC Securities Company Limited — *Shenzhen, China*

- Analyzed large-scale client transaction and portfolio data using SQL and Python to identify trends in trading pattern, driving portfolio optimization recommendations
- Performed data cleaning and standardization, improving dataset reliability for downstream analysis
- Built automated dashboards and summary tables used by 10+ analysts to monitor monthly client activity
- Automated recurring reports using Excel formulas and pivot tables, reducing manual reporting time by 8 hours/week, improving team efficiency by 30%

RESEARCH / PROJECTS

Automotive Market Modeling — May 2025 – Dec. 2025
R, Regression, Model Selection, Cross-Validation

- Built a multivariate regression pricing model in R to predict second-hand vehicle market values
- Curated and processed 150+ regional vehicle listings scraped from Craigslist to construct a clean pricing dataset
- Selected the final regression model via joint AIC, Cp, and cross-validated error minimization, achieving a test-set RMSE of 2.4 (~23% improvement over baseline)
- Optimized model specification and feature set, achieving Adjusted $R^2 = 0.79$ using core predictors such as vehicle age and odometer

NBA Game Outcome Modeling — Jan. 2025 – May 2025
R, Tidyverse, MASS, car

- Built a data extraction and preprocessing pipeline to scrape and standardize 2024–2025 NBA team-level statistics for all 30 teams
- Engineered differential performance metrics (e.g., point differential, steals) to forecast game outcomes
- Identified and removed highly collinear and leakage-prone features using Variance Inflation Factor (VIF) analysis
- Applied backward stepwise selection optimized for AIC, achieving Adjusted $R^2 \approx 0.89$ and validating model robustness

Hate Crime Predictive Analysis — May 2024 – Dec. 2024
R, Lasso Regression, EDA.

- Conducted EDA on 3,200+ NYPD hate crime incident logs (2019–2024) to analyze temporal patterns and event-driven spikes
- Engineered features from sparse categorical variables using frequency-based binning and binary encoding, transforming raw police logs into a regression-ready dataset
- Applied Lasso regression to handle high-dimensional predictors and perform feature selection on socioeconomic variables
- Built and evaluated predictive models using Lasso and AIC-based stepwise regression, achieving MAE = 1.66 and outperforming baseline daily-count forecasts

Interests*: Boxing and strength training, dance performance, independent travel, live music*